

Project Initialization and Planning Phase

Date	04 July 2024
Team ID	
Project Title	Rising Water
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report – Rising Waters (Flood Prediction Using Machine Learning)

Section	Sub-Section	Description / Details
Project Overview	Objective	Develop a machine learning-based flood prediction system that provides early and accurate flood risk predictions to help reduce damage to life, property, and infrastructure.
	Scope	Collect, preprocess, and analyze flood-related data, train machine learning models, and deploy a Flask-based web application for real-time flood prediction.
Problem Statement	Description	Existing flood warning systems often provide delayed or inaccurate predictions, making it difficult for people and authorities to prepare effectively for floods.

Section	Sub-Section	Description / Details
	Impact	Accurate flood prediction improves disaster preparedness, minimizes economic losses, protects lives and property, and supports better decision-making.
Proposed Solution	Approach	Use machine learning algorithms to analyze historical weather and flood data, predict flood risk, and display results through a user-friendly web application.
	Key Features	• Machine learning-based flood prediction model. • Real-time flood risk prediction. • User-friendly Flask web application. • Data visualization and prediction reports. • Early warning and safety recommendations.

Resource Requirement

Resource Requirements	Hardware	CPU: Intel Core i5/i7 (or equivalent) GPU: T4 GPU (optional for training) RAM: 8 GB Storage: 256 GB SSD
	Software	Programming Language: Python Framework: Flask ML Framework: Scikit-learn Libraries: Pandas, NumPy, Matplotlib, Plotly, Joblib IDE: Jupyter Notebook, Visual Studio Code
	Data	Source: Kaggle Flood Prediction Dataset Format: CSV Data: Historical flood and weather records used for training and testing the machine learning model.